

Prabhjeet Singh Sandhu

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EXPERIENCE

Delhi Police | *Technical Intern (Onsite)*

January 2026 – February 2026

- Engineered a multi-threaded Python data scraper to deploy **15 concurrent worker threads** and automated data extraction pipeline, reducing intelligence gathering time by **96%**.
- Implemented regex-based parsing algorithms to extract unstructured HTML into **3 distinct data entities** persisting structured records into an SQLite database mapped to a GUI with **automated 30-minute to 48-hour polling cycles**.

Tech Mahindra | *Software Engineer Intern (Onsite)*

September 2025 – October 2025

- Engineered an optimized **backup pipeline** utilizing Node.js Streams to dynamically extract, compress (GZIP), and securely encrypted PostgreSQL tables, bypassing V8 engine heap memory limits to enable **multi-gigabyte file processing**, reducing server memory consumption by **over 90%**.
- Automated critical database snapshot procedures using **node-cron**, eliminating **100%** of manual execution overhead and cutting backup initiation time to **under 30 seconds**.
- Designed a role-based React.js dashboard utilizing strict JWT authentication for Admin and Auditor middlewares, **securing 100% of API endpoints** and enabling teams to verify backup logs faster than manual parsing.

Valnee Solutions | *FullStack Developer Intern (Remote)*

September 2025 – November 2025

- Collaborated with a **team of 5 engineers** to architect 4 scalable full-stack applications, designing a library of **25+ reusable React components** that reduced front-end development time by 30%.
- Engineered an **AI-powered SEO** readiness tool using Google Gemini API (LLM) and Firecrawler, generating automated 0-100% performance audits for 50+ client websites.
- Spearheaded the frontend development of 'Signalmint', aggregating data across multiple REST APIs and utilizing **Chart.js** to render interactive portfolio visualizations, improving client reporting **efficiency by 40%**.

ACHIEVEMENTS

- Global Rank 158 in LeetCode Weekly Contest 477:** Placed in the top 0.5% out of 30,000+ global participants, solved **500+ DSA problems** (Leetcode, GeeksForGeeks, CodingNinjas), achieving a peak contest rating of **1933 (Leetcode Knight)**.
- Top **1,500** in nationwide **Google The Big Code 2026** and successfully solved 100% of test cases of all questions in Round 2.

PROJECTS

PrepMasterAi – Dual-Device AI Proctoring & Mock Interview Engine | ([2nd Position Hackathon](#))

[Live Demo](#) | [GitHub](#)

- Engineered a synchronized **dual-device proctoring system** using Socket.io, offloading **100% of video analysis** to the client edge to eliminate server-side processing bottlenecks and ensure **real-time anomaly detection**.
- Built a fault-tolerant audio streaming pipeline using Deepgram, implementing a **custom packet-buffering queue** to achieve **0% speech data loss** during WebSocket connection handshakes, using the Gemini/Groq-powered interview engine.
- Optimized database performance by intercepting massive Base64 proctoring images and streaming them directly to **Cloudinary**, **reducing MongoDB document payload** sizes by **over 95%** and improving recruiter dashboard efficiency.
- Architected a fully containerized microservice environment utilizing **Docker** and **Docker Compose**, automated **CI/CD** pipelines via Vercel for continuous deployment, and securely routed local network IPs to synchronize the Node.js backend.

Journeaze – AI Full-Stack Travel Booking Platform

[Live Demo](#) | [GitHub](#)

- Engineered an AI-driven travel discovery engine utilizing the **Groq API** (LLM) to parse natural language queries and dynamically generate personalized **3-day travel itineraries**, automating complex trip planning for users.
- Architected a highly optimized, **multi-parameter search interface** using React Reducers to compute 8 concurrent dynamic state constraints (price boundary algorithms, categorical filters, and rating tiers) entirely on the client side, eliminating redundant backend API calls.
- Architected a scalable MERN backend comprising **20+ RESTful endpoints**, containerizing the application utilizing **Docker** and **Docker Compose** to ensure **100% environment consistency**, while securely integrating RazorPay for transactions.

TECHNICAL SKILLS

- Languages:** C++ (Proficient), C, Python, JavaScript, SQL
- Frontend Development:** React.js, Next.js, Context API, React Reducers, HTML5, CSS3, Tailwind CSS
- Backend & Database:** Node.js, Express.js, MongoDB, MySQL, PostgreSQL, Mongoose, RESTful APIs, NoSQL
- Architecture & Security:** Real-Time (WebSockets, Socket.io, WebRTC), JWT Cookies, Role-Based Access Control (RBAC)
- AI & Data Science:** LLM Integration (Google Gemini, Groq API), Supervised Machine Learning, Pandas, NumPy, Scikit-Learn
- Tools & Infrastructure:** Git, GitHub, Docker, CI/CD, Vercel, Postman, Cloudinary, Deepgram SDK, RazorPay API

EDUCATION

Guru Tegh Bahadur Institute of Technology (GGSIUP)

September 2023 – September 2027

B.Tech in Computer Science and Engineering | CGPA: 9.5

- Coursework:** Data Structures & Algorithms (DSA), Object Oriented Programming (OOP), Multithreading, Concurrency, Complexity Analysis, Operating Systems, Computer Networks, DBMS, System Design.